

Figure5B

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Intro

Compares the bNTI values between semesters to reproduce the results of Figure 5B from the original publication Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
```

2. Load data

bNTI_weighted_26_1000_asvs.csv weighted beta Nearest Taxin Index (NTI) values for the 1000 most abundant ASVs in the dataset. samo_metadata_workable.tsv contains the metadata for the samples. date2season2community.tsv is a table mapping the date, season, and community columns.

```
bNTI <- read_csv("../data/tree_analysis/bNTI_weighted_26_1000_asvs.csv", show_col_types = FALSE,
                 col_names = T) %>% column_to_rownames("...1")
```

```
## New names:
## * `` -> `...1`
```

```
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUNITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Extend metadata table

```
METADATA_ext <- METADATA %>%
  left_join(DATE2SEASON2COMMUNITY,
            by =c("Date"))
```

4. Convert bNTI to long format

```
bNTI_long <- bNTI %>%
  as.data.frame() %>%
  rownames_to_column("sample_name") %>%
  pivot_longer(cols = -sample_name,
               names_to = "sample_name_vs",
               values_to = "bnti") %>%
  filter(is.na(bnti) != T)
```

5. Select samples

```
sample2date <- METADATA_ext %>%
  select(Date, sample_name)

bNTI_long_sel <- bNTI_long %>%
  filter(sample_name %in% sample2date$sample_name &
         sample_name_vs %in% sample2date$sample_name)
```

6. Add metadata

```
METADATA_vars <- METADATA_ext %>% select(sample_name, Date, Community)

bNTI_long_sel_ext <- bNTI_long_sel %>%
  left_join(METADATA_vars, by = c("sample_name" = "sample_name")) %>%
  left_join(METADATA_vars, by = c("sample_name_vs" = "sample_name"), suffix = c("", "_vs"))
  mutate(type = paste(Community, Community_vs, sep = "_vs_"),
         type2 = if_else(Community == Community_vs & Community == "S1", "within_S1", "between_S1"),
         type2 = if_else(Community == Community_vs & Community == "S2", "within_S2", "between_S2"),
         type2 = if_else(Community != Community_vs, "btween", type2))
```

7. Select and order dates from S1 and S2

```
s1_dates_ord <- METADATA_vars %>%
  filter(Community == "S1") %>%
  select(Date) %>%
  arrange(Date)

s2_dates_ord <- METADATA_vars %>%
  filter(Community == "S2") %>%
  select(Date) %>%
  arrange(Date)
```

8. Select consecutive dates from S1

```
bNTI_long_sel_ext_s1_list <- list()

for (i in 2:dim(s1_dates_ord)[1]) {

  date_i_curr <- s1_dates_ord[i, "Date", drop = T]
  date_i_prev <- s1_dates_ord[i-1, "Date", drop = T]
  dates_i <- c(date_i_curr, date_i_prev)
  dates_i_str <- paste(dates_i[1], dates_i[2])

  bNTI_long_sel_ext_s1_list[[dates_i_str]] <- bNTI_long_sel_ext %>%
    filter(as.character(Date) %in% c(dates_i) &
           as.character(Date_vs) %in% c(dates_i))

}

# create a data frame from all elements in the list
bNTI_long_sel_ext_s1 <- do.call(rbind, bNTI_long_sel_ext_s1_list)
```

9. Select consecutive dates from S2

```
bNTI_long_sel_ext_s2_list <- list()

for (i in 2:dim(s2_dates_ord)[1]) {

  date_i_curr <- s2_dates_ord[i,"Date", drop =T]
  date_i_prev <- s2_dates_ord[i-1,"Date",drop =T]
  dates_i <- c(date_i_curr, date_i_prev)
  dates_i_str <- paste(dates_i[1], dates_i[2])

  bNTI_long_sel_ext_s2_list[[dates_i_str]] <- bNTI_long_sel_ext %>%
    filter(as.character(Date) %in% c(dates_i) &
           as.character(Date_vs) %in% c(dates_i))

}

bNTI_long_sel_ext_s2 <- do.call(rbind, bNTI_long_sel_ext_s2_list)
```

10. Concat both data frames

```
bNTI_long_sel_ext_ord <- rbind(bNTI_long_sel_ext_s1, bNTI_long_sel_ext_s2)
```

```
bNTI_percs <- bNTI_long_sel_ext_ord %>%
  group_by(type) %>%
  summarise(n = n(),
            n_bnti_gt_2 = sum(bnti > 2),
            n_bnti_lt_2 = sum(bnti < -2),
            perc_bnti_gt_2 = n_bnti_gt_2/n,
            perc_bnti_lt_2 = n_bnti_lt_2/n,
            max_bnti = max(bnti)) %>%
  arrange(desc(n))
```

11. Get percentage of samples with bNTI greater or lower than 2

12. Plot bNTI per transition type

```
text_size <- 10

bnti_boxplots <- ggplot(bNTI_long_sel_ext_ord,
  aes(x = type, y = bnti,
      fill = type)) +

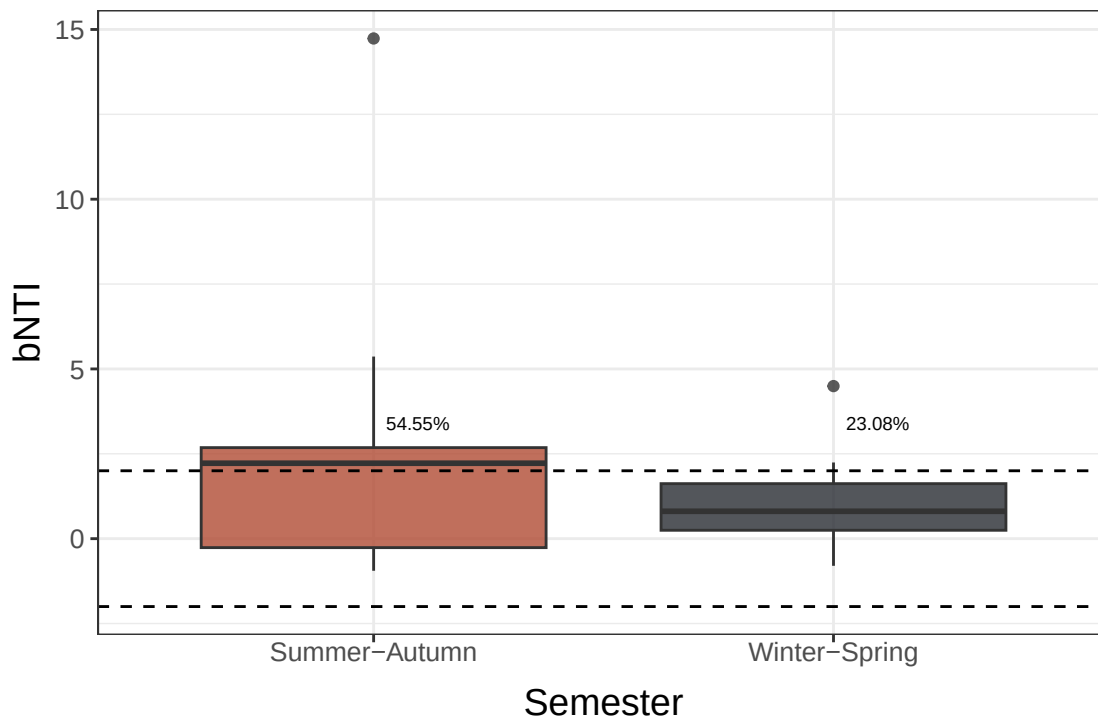
  xlab("Semester") +
  ylab("bNTI") +
  geom_boxplot(alpha = 0.8) +
  geom_hline(yintercept = 2, linetype = "dashed") +
  geom_hline(yintercept = -2, linetype = "dashed") +
  scale_fill_manual(values = c("#b14b34", "#292c33")) +
  scale_x_discrete(labels = c("S1_vs_S1" = "Summer-Autumn",
                              "S2_vs_S2" = "Winter-Spring")) +

  theme_bw() +
```

```

theme(
  axis.text.x = element_text(size = text_size, margin = margin(b = 5)),
  axis.text.y = element_text(size = text_size),
  axis.title = element_text(size = text_size + 4),
  strip.text = element_text(size = text_size + 4),
  strip.background = element_blank(),
  # extend margin
  plot.margin = margin(1, 1, 1, 1, "cm")
) +
# add percentage of samples with bnti > 2 above bar plot
# move the number for better visualization
geom_text(data = bNTI_percs, aes(x = type, y = 3,
                                label = scales::percent(perc_bnti_gt_2, accuracy = 0.01),
                                hjust = -0.2),
          vjust = -0.5, size = 2.5) +
guides(fill = "none")
bntni_boxplots

```



13. Print session info

```

sessionInfo()

## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS

```

```

##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [6] LC_MESSAGES=en_US.UTF-8 LC_PAPER=en_US.UTF-8 LC_NAME=C LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] fANCOVA_0.6-1 maps_3.4.2 corrgram_1.14 ggpubr_0.6.0 ggcorrplot_0.1.4.1
## [8] permute_0.9-7 lubridate_1.9.3 forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4
## [15] tidyr_1.3.1 tibble_3.2.1 ggplot2_3.5.1 tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.5 xfun_0.48 rstatix_0.7.2 tzdb_0.4.0 vctrs_0.6.5 tools_
## [9] fansi_1.0.6 highr_0.11 cluster_2.1.6 pkgconfig_2.0.3 Matrix_1.7-0 lifecycle_
## [17] tinytex_0.53 munsell_0.5.1 carData_3.0-5 htmltools_0.5.8.1 yaml_2.3.10 Formula_
## [25] car_3.1-3 MASS_7.3-61 abind_1.4-8 nlme_3.1-166 tidyselect_1.2.1 digest_
## [33] splines_4.4.2 cowplot_1.1.3 fastmap_1.2.0 colorspace_2.1-1 cli_3.6.3 magrittr_
## [41] withr_3.0.1 scales_1.3.0 backports_1.5.0 bit64_4.5.2 timechange_0.3.0 rmarkdown_
## [49] hms_1.1.3 evaluate_1.0.1 knitr_1.48 mgcv_1.9-1 rlang_1.1.4 glue_
## [57] R6_2.5.1

```